Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/17/2006
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Project Title:	Temperature and Composition Dependence of Mass and Thermal Diffusion in Liquid Metals and Compound Semiconductor Alloys		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35899	**Congressional District:**	5
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Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alexander, Iwan (Case Western Reserve University)		
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	The goal of this project is to obtain a quantitative understanding of the temperature and concentration dependence of liquid-state mass and thermal transport properties of selected semiconductors, to develop data that will facilitate investigation of their liquid structures, and to obtain benchmark values; ultimately enabling better prediction of the solidification behavior. Selected group IV, III-V, and II-VI semiconductors are chosen based on their technological importance and the range of predicted (but as yet unsubstantiated) temperature and concentration dependences. The solidification of elemental semiconductors is reproducible on an industrial scale. However, for binary group IV and III-V compounds mesoscale solidification patterns are still not predicable. This problem is severe in the II-VI systems where the yield of device quality material varies considerably between solidification runs and this limits their ultimate commercial utility; i.e., yields of 5-10% for active devices is considered excellent. Better control of the mesoscale		

Task Description:	structure is required for smaller devices. There is clearly much room for improvement. A better understanding of the interplay of mass and thermal transport would aid these improvements. However, a “trial and error” approach is costly, time consuming and, may be the result of a fortuitous cancellation of errors. One way is to simulate the key steps in the processing of such materials. Indeed, advances in our understanding and control of materials preparation for devices has benefited from numerical modeling efforts. In particular, insight into bulk transport and segregation processes has been obtained. However, for these modeling efforts to yield further guidance, and consequently, improvements in device yields, accurate data for molten state diffusivities and thermophysical properties, and, in particular, their temperature dependence, is essential. At present such data is either missing or poorly defined. Experimentally determined transport properties, such as mass and thermal diffusivity, are easily contaminated by buoyancy-driven convective effects. The accurate determination of transport properties requires convective flow velocities to be eliminated or significantly suppressed. This is generally not possible under terrestrial conditions. It is often argued that mass diffusivity measurements techniques such as NMR and Inelastic Neutron Scattering that probe rapid molecular motion are insensitive to convective contamination; this is erroneous. Furthermore, to extract a diffusivity value from such data depends critically upon assumptions made about the structure of the liquid in which the measurements are made. Macroscopic techniques such as the ones proposed here, while prone to convective contamination require no assumptions about the liquid’s structure. Thus, provided that convective contamination can be eliminated or suppressed to acceptable levels, macroscopic techniques promise a more precise measurement. A number of methods to reduce convective contamination have been attempted. The use of narrow capillaries to suppress convective transport has met with limited success. In addition, arguments have been made (to explain anomalous data) that narrow capillaries produce “wall effects” that locally modify the diffusion coefficient. Magnetic fields, widely used for suppressing convection in conducting liquids, may modify the diffusive motion itself. Earlier liquid (metal) diffusion studies conducted on spacecraft have demonstrated the gain in precision resulting from the drastic reduction of convection in a low-gravity environment. However, in general, these experiments have been so limited and conducted in such specific systems that we are unable to unambiguously interpret the temperature and composition dependences of self-diffusivity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress provided this reporting period.
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